

PAULATUK, NT, Canada

WMO: 719840

Lat: 69.360N

Lon: 124.080W

Elev: 6

StdP: 101.25

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-37.4	-35.8	-41.5	0.1	-36.7	-39.6	0.1	-35.6	21.4	-17.5	19.1	-17.6	2.4	240	0.992

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	22.0	14.9	19.3	13.8	16.9	12.7	15.6	20.9	14.3	18.6	12.9	16.5	5.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	9.4	17.6	11.9	8.7	16.4	10.7	8.0	15.0	43.9	21.3	40.1	18.5	36.5	16.6	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.8	13.0	11.1	-39.4	26.3	2.7	2.4	-41.3	28.1	-42.9	29.5	-44.5	30.9	-46.5	32.7		
			-39.4	17.8	2.8	1.8	-41.5	19.1	-43.1	20.2	-44.7	21.2	-46.8	22.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.6	-24.5	-24.7	-23.5	-14.3	-3.0	5.4	10.3	8.8	3.7	-5.2	-15.1	-22.1	(6)
	DBStd	14.28	6.69	6.76	7.34	7.27	5.94	4.53	4.51	4.28	4.12	4.62	5.69	6.53	(7)
	HDD10.0	6904	1068	972	1039	728	402	154	52	76	193	472	752	997	(8)
	HDD18.3	9828	1327	1205	1297	978	660	387	251	297	439	730	1002	1255	(9)
	CDD10.0	122	0	0	0	0	1	17	62	38	4	0	0	0	(10)
	CDD18.3	6	0	0	0	0	0	1	3	2	0	0	0	0	(11)
	CDH23.3	37	0	0	0	0	0	8	19	9	0	0	0	0	(12)
	CDH26.7	4	0	0	0	0	0	1	2	1	0	0	0	0	(13)
(14) Wind	WSAvg	4.5	5.0	4.9	4.5	4.0	4.0	3.6	3.9	4.3	4.8	5.0	5.1	4.6	(14)
(15) Precipitation	PrecAvg	219	11	11	15	14	12	17	25	37	24	21	14	12	(15)
	PrecMax	349	24	21	64	95	47	48	42	83	65	73	61	40	(16)
	PrecMin	144	0	0	0	0	0	3	9	14	1	2	1	1	(17)
	PrecStd	64	6	7	14	18	11	12	11	17	13	15	12	9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.1	-6.4	-2.5	6.1	13.9	23.7	26.0	24.6	17.1	7.8	0.9	-4.9	(19)
		MCWB	-3.8	-7.2	-3.7	3.3	8.5	15.6	17.1	16.3	12.2	5.8	-0.3	-5.5	(20)
	2%	DB	-7.9	-9.3	-6.0	2.1	9.6	19.5	22.7	20.5	13.8	3.1	-2.3	-7.8	(21)
		MCWB	-8.6	-10.1	-7.1	0.3	5.8	13.1	15.1	14.6	10.0	1.9	-3.0	-8.4	(22)
	5%	DB	-11.1	-11.4	-9.2	-0.6	6.9	16.6	20.1	17.7	11.3	1.6	-5.1	-9.8	(23)
		MCWB	-11.6	-11.9	-9.8	-1.9	3.9	11.6	14.4	13.4	8.7	0.7	-5.7	-10.3	(24)
	10%	DB	-14.6	-14.0	-12.7	-3.3	4.6	13.4	17.7	15.1	9.3	0.4	-7.1	-12.3	(25)
		MCWB	-15.0	-14.4	-13.3	-4.4	2.4	9.6	13.5	12.0	7.5	-0.4	-7.7	-12.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.9	-7.0	-3.6	3.3	8.3	15.8	17.8	17.1	12.5	5.8	-0.2	-5.4	(27)
		MCDB	-3.1	-6.4	-2.6	6.0	13.0	23.5	24.0	23.4	16.8	7.7	0.9	-4.9	(28)
	2%	WB	-8.4	-9.9	-6.9	0.2	5.9	13.5	16.1	15.0	10.3	2.2	-3.1	-8.2	(29)
		MCDB	-7.8	-9.3	-5.9	1.6	9.3	18.8	21.4	19.7	13.2	3.0	-2.5	-7.7	(30)
	5%	WB	-11.6	-11.9	-9.8	-2.0	4.0	11.5	14.9	13.6	8.8	0.7	-5.7	-10.2	(31)
		MCDB	-11.0	-11.3	-9.1	-0.8	6.6	16.2	19.2	17.3	10.8	1.4	-5.2	-9.8	(32)
	10%	WB	-15.0	-14.4	-13.1	-4.5	2.4	9.6	13.6	12.3	7.7	-0.3	-7.7	-12.6	(33)
		MCDB	-14.6	-13.9	-12.5	-3.4	4.3	13.3	17.5	14.9	9.4	0.4	-7.1	-12.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.8	7.5	8.6	7.0	7.5	7.0	5.8	4.5	4.3	5.7	6.3	(35)
		MCDBR	10.0	8.8	9.4	9.1	9.3	14.5	11.4	10.1	7.7	4.2	5.8	8.9	(36)
	5% WB	MCWBR	9.6	8.3	8.7	8.3	6.4	9.0	6.3	5.5	5.0	3.7	5.7	8.7	(37)
		MCWBR	10.1	8.6	9.4	9.4	9.1	13.9	10.6	9.2	7.2	4.1	5.8	9.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.155	0.207	0.249	0.303	0.335	0.361	0.351	0.335	0.300	0.230	0.156	N/A	(40)
		taud	1.899	2.072	2.211	2.127	2.181	2.313	2.432	2.499	2.515	2.176	1.913	N/A	(41)
	Ebn at Noon		52	572	786	828	842	830	830	805	739	521	37	0	(42)
		Edh at Noon	4	40	71	108	117	106	92	77	57	35	3	0	(43)
(44) All-Sky Solar Radiation	RadAvg		0.02	0.42	2.00	4.44	6.08	6.16	5.17	3.27	1.69	0.49	0.06	0.00	(44)
	RadStd		0.00	0.05	0.08	0.20	0.37	0.45	0.45	0.38	0.18	0.06	0.01	0.00	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+1.35	+1.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.56	+1.18	+1.54	N/A	N/A	N/A	-221	-215	N/A	N/A

Nomenclature: See separate page